

ISM: Mag:

$$I_i = \frac{L_i}{4\pi D_i^2}$$

$$\frac{I_1}{I_2} = (100)^{\frac{2}{5}(M_2 - M_1)}$$

Distance modulus: $I_1 = \frac{L_{B/V}}{4\pi D^2}$ (apparent)
(absolute) $I_2 = \frac{L_{B/V}}{4\pi 10^2}$

$$* \frac{L (10)^2}{(L_{B/V}) D^2} = \left(\frac{10}{D}\right)^2 = 100^{\frac{1}{5}(M - m)}$$

$$\log\left(\left(\frac{10}{D}\right)^2\right) = \frac{2}{5}(M - m)$$

$$2 - 2 \log(D) = \frac{2}{5}(M - m)$$

$$5 - 5 \log(D) = M - m$$

or

$$m - M = 5 \log(D) - 5$$

Extinction:

$$m_{\text{obs}} - m_{\text{true}} = A_\lambda$$

$$A_\lambda = 1.09 \tau_\lambda \leftarrow \text{optical depth}$$

* Differential extinction

$$E(\lambda - V) = A_\lambda - A_V \leftarrow V/B - \text{bands}$$

$$E(\lambda - B) = A_\lambda - A_B$$

$$A_{\lambda - B} = \frac{A_V}{E(B - V)} = \frac{A_V}{A_\lambda - A_V}$$

$$R = \frac{\frac{1}{\lambda_V^4}}{\frac{1}{\lambda_B^4}} - 1 = 0.596$$

(B.1) $A_V = 0.65 \text{ mag}$, $A_\lambda \propto \lambda^{-1.2}$

$$A_B - A_V = A_B \left(1 - \frac{A_V}{A_B} \right)$$

$$A_B - A_V = A_B \left(1 - \frac{\lambda_B^{1.2}}{\lambda_V^{1.2}} \right)$$

$\propto$

$$A_B - \alpha A_B = A_V$$

$$A_B = A_V / (1 - \alpha)$$

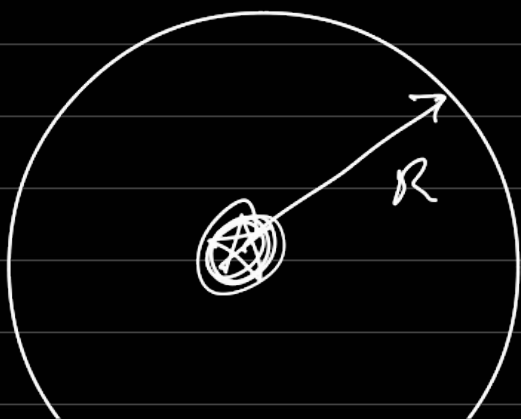
$$\frac{I_1}{I_2} = 10^{\frac{2}{5}(M_\odot - M_B)} = \frac{L_B}{L_\odot}$$

$$\log \left(\frac{L_B}{L_\odot} \right) = \frac{2}{5} (M_\odot - M_B)$$

$$M_\odot - M_B = \frac{2}{5} A_B$$

$$\log \left(\frac{L_B}{L_\odot} \right) = A_B$$

$$L_B = L_\odot e^{-\frac{2}{5} A_B}$$



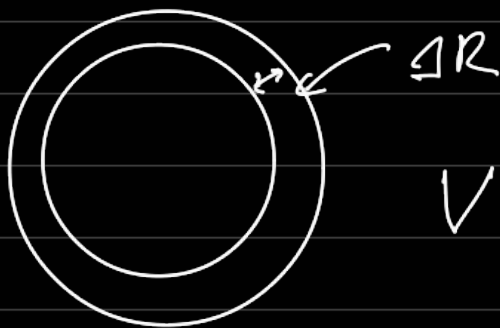
(i) Ionization: $[S^*] = \frac{\# \text{ photons which will ionize}}{\Delta t}$

(ii) Recomb: $R \propto n_p$

$$[R] = \frac{\# \text{ of recomb}}{\Delta t}$$

(i) $\underline{N} = S^* \Delta t$

$$\frac{N}{V_{\text{vol}}} = n_e$$



$$V = 4\pi R^2 \Delta R$$

$$N = n_e 4\pi R^2 \Delta R = S^* \Delta t$$

$\Rightarrow$

(ii) $R = \alpha n_p$ per electron

$$R_{\text{tot}} = N_e (\alpha n_p) \\ = \frac{4\pi}{3} R^3 n_e \propto n_p$$

$$R_{\text{tot}} = S^* = \frac{4\pi}{3} R_s^3 n_e \propto n_p$$

$$R_s = \left(\frac{3S^*}{4\pi n_e \alpha n_p} \right)^{1/3}$$

$\uparrow n_p^2$

$$n_e = n_p$$

③ $R_{\text{tot}} = \frac{N}{\Delta t} = S^*$

$\uparrow \Delta t$

$$\Rightarrow E_{\lambda} S_{\lambda} \approx \frac{h\nu}{\Delta t} = L$$

$$L = \left(\frac{hc}{\lambda} \right) S_{\lambda}$$

* Oxygen collisional excitation:

- Neutral H:
- OII (O^+), OIII (O^{++}), N^+ NII
 $\rightarrow$ electrons in ISM are "cold" - non-relativistic

$$\bullet T \sim 10^3 K \quad SII, Si, CII$$

$$\bullet T \sim 28,000 K \quad Ly\alpha$$

$$T \sim 10^4 K \quad OII, OIII, NII$$

* Collisions ^{radiative} $\rightarrow$ tells us n_e

$\lambda 3727$ OII (forbidden transition)

